

MUHAMMED SHAMEEL

Data Scientist | Machine Learning Engineer

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Summary

MSc Data Science and Business Analytics graduate with a Mathematics background and hands-on experience in Machine Learning, Retrieval-Augmented Generation (RAG), and AI-powered application development. Proficient in Python, FastAPI, Scikit-learn, SQL, statistical analysis, and feature engineering, with experience building end-to-end machine learning solutions, data pipelines, and cloud-deployed applications. Passionate about developing scalable AI systems that solve real-world problems.

TECHNICAL EXPERTISE

Programming Language

Python, SAS, SQL

Data Engineering

Data Cleaning, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Semantic Chunking, Embedding Generation, Metadata Extraction, Vector Databases

Data Visualization & Tools

Matplotlib, Tableau, SPSS, Jupyter Notebook, VS Code

Machine Learning & AI

Scikit-learn, TensorFlow, Keras, Random Forest, XGBoost, Support Vector Machines (SVM), Neural Networks, Model Evaluation, Hyperparameter Tuning, Retrieval-Augmented Generation (RAG), Prompt Engineering

Statistical Analysis

Hypothesis Testing, Regression Analysis, ANOVA, Correlation Analysis, Factor Analysis, Feature Engineering

Backend & Cloud Computing

FastAPI, API Integration, JSON, Git, GitHub, Docker, Render, Vercel, AWS Fundamentals

Work Experience

Software Engineering Intern

05/2026 – Present | Kerala, India

InfoCreon Solutions Private Limited

- Contributed to the design, development, and deployment of AI-driven software solutions in a collaborative engineering environment.
- Developed backend services using **Python** and **FastAPI**, supporting data processing, machine learning integration, and application functionality.
- Applied data preprocessing, feature engineering, statistical analysis, and model evaluation techniques to improve AI-driven solutions.
- Participated in cloud deployment, version control, testing, debugging, and iterative software improvements using modern development practices.
- Worked with cross-functional technologies including Docker, Git, Render, and Vercel while leveraging AI-assisted development tools to improve productivity.

Mathematics Tutor

07/2023 – 03/2024

- Delivered one-on-one mathematics tutoring to undergraduate and high-school students.
- Improved student understanding and academic performance through personalized learning strategies.

Projects

Grocery AI – Retrieval-Augmented Generation (RAG)

07/2026 – Present

github.com/Muhammed-Shameel/Grocery-AI

- Developed an end to end Retrieval-Augmented Generation (RAG) application to provide context-aware answers from grocery-related knowledge sources.
- Implemented semantic chunking, metadata enrichment, and embedding generation to improve retrieval quality and contextual understanding.
- Built a document processing pipeline supporting structured knowledge extraction and vector-based semantic search.
- Integrated FastAPI services to support scalable retrieval workflows and LLM-ready responses.
- Designed the project with an extensible architecture for future automated data ingestion and knowledge base expansion.

Tech Stack: Python, FastAPI, Sentence Transformers, Vector Embeddings, **RAG**, **Semantic Chunking**

FLUVIO-AI-Powered Flood Intelligence Dashboard

05/2026 – 07/2026

flood-analysis-dashboard.vercel.app/

github.com/Muhammed-Shameel/Flood-Intelligence-Dashboard

- Developed an AI-powered flood intelligence dashboard integrating machine learning predictions with interactive geospatial visualization and real-time analytics.
- Built a FastAPI backend for machine learning inference, data processing, and analytics APIs powering the dashboard.
- Performed data preprocessing and feature engineering to improve flood risk prediction using environmental and geospatial data.
- Designed an interactive analytics interface with region-based filtering, KPI reporting, historical and live data visualization, and export functionality.
- Deployed the backend on Render and frontend on Vercel using Docker and Git-based deployment workflows.

Tech Stack: Python, **FastAPI**, Scikit-learn, Next.js, TypeScript, Docker, Render, Vercel, Git

A Data-Driven Analysis on Football Players Under Different Managers,

github.com/Muhammed-Shameel/football-player-analysis

- Analyzed **2,500+** match records spanning multiple Premier League seasons.
- Applied **Factor Analysis** and feature engineering to derive latent player performance metrics.
- Evaluated multiple machine learning models, achieving **68.7% accuracy** and **0.65 macro F1-score** with XGBoost.
- Generated tactical insights into manager-player compatibility and performance adaptation.

Tech Stack : Python (Pandas, NumPy, Scikit-learn), SQL, Matplotlib, Seaborn, Flask

CSIRO Image2Biomass Prediction,

github.com/Muhammed-Shameel/CSIRO-Biomass-Prediction-using-Machine-Learning

- Developed regression models using MSE, Huber, and LogCosh loss functions.
- Achieved **$R^2 = 0.86$** through ensemble learning and hyperparameter optimization.
- Built a weighted ensemble model to improve prediction stability across heterogeneous data.

Tech Stack : Python, TensorFlow, Keras, NumPy, Matplotlib

Education

Asia Pacific University of Technology & Innovation, Malaysia

03/2026

Master of Science in Data Science and Business Analytics

GPA: 3.62

Kottakkal Farook Arts and Science College (Affiliated under University of Calicut)

04/2024

Bachelor of Science in Mathematics